SECTION B

Answer **all** questions.

8. Read through the following article carefully. Paragraph

Some Interesting Aspects of Quantum Physics

1 The whole idea of quantum physics usually starts with the idea that the energy of photons is quantised. Following the work of Planck and Einstein, the smallest packet of electromagnetic energy (the photon) has an energy given by:

$$E = hf \quad \textbf{Equation 1}$$

2 This simple little equation is no stranger to A level physics students but within six years of its inception it had solved the problems of black body radiation (by not allowing too many high energy photons) and the photoelectric effect (by explaining why the kinetic energy of photoelectrons depends on the wavelength of light and not its intensity).

3 In 1913, quantum physics took another enormous leap when Niels Bohr applied it to electron orbitals in atoms. In his new theory, the energy levels of electrons are quantised and electromagnetic radiation is emitted or absorbed when electrons make jumps from one quantised level to another. In a later interpretation of the Bohr theory, the circumference of any electron orbit must be a whole number of electron wavelengths (so that the electron's de Broglie wavelength leads to a stationary wave).

$$n\frac{h}{p} = 2\pi r \quad \rightarrow \quad r = \frac{nh}{2\pi p} \quad \textbf{Equation 2}$$

4 where n is an integer, p is the momentum of the electron and r the radius of its orbit. If we are concerned with the hydrogen atom, equating the electrostatic force between the proton and electron with the centripetal force leads to Equation 3 (m_e and e are the mass and charge of an electron and ε_0 the permittivity of free space).

$$\frac{m_e v^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2} \quad \rightarrow \quad p^2 = \frac{m_e e^2}{4\pi\varepsilon_0 r} \quad \textbf{Equation 3}$$

5 Substituting for r in Equation 3 from Equation 2 gives:

$$p = \frac{m_e e^2}{2\varepsilon_0 nh} \quad \textbf{Equation 4}$$

and the starting point of Equation 3 can also be used to show that the kinetic energy of the electron in its orbit is given by:

$$\frac{1}{2}m_e v^2 = \frac{e^2}{8\pi\varepsilon_0 r} \quad \textbf{Equation 5}$$

However, the electrostatic potential energy of the electron and proton in hydrogen is: 6

$$\mathrm{PE} = -\frac{e^2}{4\pi\varepsilon_0 r} \quad \textbf{Equation 6}$$

so that the total energy of the hydrogen atom is:

$$\text{Total energy} = \mathrm{PE} + \mathrm{KE} = -\frac{e^2}{4\pi\varepsilon_0 r} + \frac{e^2}{8\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r} \quad \textbf{Equation 7}$$

Using Equations 4 and 7 and the rather useful relationship: 7

$$\mathrm{KE} = \frac{p^2}{2m} \quad \textbf{Equation 8}$$

it is reasonably straightforward to show that the total energy of the hydrogen atom is:

$$\text{Total energy (in eV)} = -\frac{p^2}{2m} = -\frac{m_e e^3}{8\varepsilon_0{}^2 n^2 h^2} \quad \textbf{Equation 9}$$

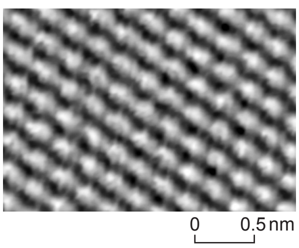
It turns out that when the integer n = 1, we have the ground state of hydrogen and n = 2
corresponds to the first excited state etc. By inputting n = 1 into Equation 9, we find that 8
the energy of the ground state of hydrogen is approximately -13.6 eV. Hence, the total
energy of the n^{th} level of hydrogen (in eV) is given by Equation 10.

$$\text{Total energy (in eV)} = -\frac{13.6}{n^2} \quad \textbf{Equation 10}$$

The success and accuracy of Niels Bohr's theory of the hydrogen atom was almost
unbelievable and meant that quantum physics was well and truly established as the main 9
research topic in science for the next century.

When, in 1926, Erwin Schrödinger developed the quantum mechanical wave equation,
there was another explosion in quantum physics research. Schrödinger's publication of 10
1926 included a complete (non-relativistic) solution of the hydrogen atom which is the
foundation of the whole subject of chemistry!

Another highlight directly obtainable from his famous wave equation is the concept of
quantum mechanical tunnelling – where particles can tunnel through energy barriers
that are too high for them. This is the explanation for alpha particles escaping from large
nuclei in the process of alpha decay. The strong nuclear force is so great that the energy 11
required for an alpha particle to escape a nucleus is far too big and
no alpha particles should ever be detected. However, this tunnelling
effect means that there is a small but finite chance that alpha
particles can, indeed, escape.



This tunnelling effect has also led to microscopes that can "see"
individual atoms e.g. the graphite atoms in the hexagonal pattern in 12
the image.

Examiner only

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) (i) Explain how Einstein's photoelectric equation arises from Equation 1 ($E = hf$) and the principle of conservation of energy (see Paragraphs 1 and 2). [2]

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(ii) The intensity of light does not affect the maximum kinetic energy of photoelectrons emitted from the cathode of a photocell. State what electrical quantity is proportional to the intensity. [1]

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(b) Show how the substitution for r in Equation 3 $\left(p^2 = \dfrac{m_e e^2}{4\pi\varepsilon_0 r}\right)$ using Equation 2 $\left(r = \dfrac{nh}{2\pi p}\right)$ gives Equation 4 $\left(p = \dfrac{m_e e^2}{2\varepsilon_0 nh}\right)$. [2]

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Examiner only

(c) (i) Use Equation 9 $\left(\text{Total energy (in eV)} = -\dfrac{m_e e^3}{8\varepsilon_0{}^2 n^2 h^2}\right)$ to show that the energy of the n = 1 state of hydrogen is −13.6 eV. [2]

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(ii) Use Equation 10 $\left(\text{Total energy (in eV)} = -\dfrac{13.6}{n^2}\right)$ to explain why the ionisation energy of hydrogen is 13.6 eV. [2]

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(d) A student suggests that the hydrogen 656 nm red line is emitted from a hydrogen atom when an electron drops from the second excited level (n = 3) to the first excited level (n = 2). Determine whether or not this is true.

Use Equation 10 $\left(\text{Total energy (in eV)} = -\dfrac{13.6}{n^2}\right)$. [3]

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Examiner only

(e) Explain how quantum tunnelling will provide a lower than expected value for V (see Equation 11 $\left(eV = \frac{hc}{\lambda}\right)$ and Paragraph 14). [2]

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(f) (i) Explain why an increased temperature provides a greater current in a LED (see Paragraph 15). [2]

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(ii) Calculate a typical thermal kinetic energy of a particle at room temperature in eV and use this to explain why it is difficult to measure the "switch-on" voltage of a diode accurately. [4]

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END OF PAPER